

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 2

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GitHub Repository Link: [https://github.com/LilyTheGreat101/-lab-2-](https://github.com/LilyTheGreat101/-lab-2-predictive-analytics)

[predictive-analytics](#)

Date Submitted: 3rd July 2026

1. AI Use Summary

Did you use any AI tool for this lab? :Yes

Estimated percentage of the work influenced by AI :20%

Did you attach evidence of AI use where applicable? :Yes

2. Details of AI Use

| Name of AI Tool Used | Purpose for Using the Tool | Prompt or Instruction Given to the Tool | Part(s) of the Work Influenced | How I Verified, Edited, or Improved the Output |

| ChatGPT | Debugging and fixing search algorithm implementations | "why is my code not running"
| I ran all code in my notebook, fixed errors step-by-step, tested assertions, and adjusted code based on lab requirements |

ChatGPT | Understanding AI search concepts and heuristics | "simple explanation of Clusters| Model Training", "Data cleaning" | Reflection questions (Section 16) | I made sure I understood the concepts and applied my thinking |

ChatGPT | Collab and github linking guidance | "how to push code to GitHub repo" | README.md and repository setup, final checklist | I edited README to match my project and verified Git commands on my system |

3. Attachment of AI Output Evidence

| AI-generated draft | No

| Prompt history | No

| Screenshot of AI interaction | Yes | screenshots.png |

| Exported AI conversation | No

| AI-generated code snippet | No

| Revised version showing student input | No

4. Student Declaration

I declare that:

| The submitted work is my own work. | Yes |

| Any use of AI tools has been clearly declared in this form. | Yes |

| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes |

| I reviewed, tested, edited, and improved any AI-generated content before submission. | Yes |

| My AI usage does not exceed 25% of the entire work. | Yes |

| I understand that undeclared or excessive AI use may be treated as academic misconduct. | Yes |

Student Signature:Lily Great Ayensu

Date:3rd July 2026

AI Lab Overview

ChatGPT

Split data

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Show more

Alright I see exactly why your code is "not working" — and yeah this is a classic trap.

I'm gonna break it like a lecturer marking scripts.

WHAT'S WRONG IN YOUR CODE

1. You scaled BEFORE splitting (BIG ERROR)

Python

X_scaled = scaler.fit_transform(X)

Run

Why this is wrong:

You are using ALL the data (including test set) to fit the scaler.

That's called data leakage

Meaning:

your model "peeks" at test data distribution

results become fake-good / unreliable

2. You destroyed your proper splits

You did:

Python

X_train = X_scaled[1:len(X_train)]

X_test = X_scaled[1:len(X_train):]

Run

+ Ask anything

ChatGPT can make mistakes. Check important info.

AI Lab Overview

ChatGPT

Create a public repository called lab-2-predictive-analytics on GitHub/GitLab.

Clone it to your machine (or link it to Google Colab).

Create the notebook lab_2_predictive_analytics.ipynb inside the repo (a scaffolded template is provided).

they said tolink sum

Yeah — this is that "GitHub linking" step they expect, and it's easy marks if you do it clean.

I'll show you exactly what they mean in real terms.

What they ACTUALLY want

They are asking you to:

1. Create a GitHub repo

Name:

lab-2-predictive-analytics

2. Link it to your notebook (Colab OR local)

This is the part you're asking about.

You have 2 correct options:

OPTION A (EASIEST) — Link Colab to GitHub

Do this:

Step 1 — Open your notebook in Colab

Step 2 — Click:

File → Save a copy in GitHub

+ Ask anything

ChatGPT can make mistakes. Check important info.

AI Lab Overview

ChatGPT

ah ok i did the linking

UpgradeShare

Good — that's one of the submission boxes ticked.

Now don't just stop there like the job is done 🧠 we still need final submission sanity check so you don't lose free marks.

FINAL CHECKLIST (DO THIS BEFORE YOU SUBMIT)

1. Notebook runs clean

Go to:

Runtime → Restart and run all

✓ No red errors anywhere

✓ Every cell runs top-to-bottom

2. File name is correct

Must be EXACT:

lab_2_predictive_analytics.ipynb

(Not "final", not "copy", not "version2")

3. Sections exist exactly like lab

You should have:

Section 1

- 1.1 Load & Explore
- 1.2 Preprocessing & Feature Engineering
- 1.3 Split
- 1.4 Model + Overfitting

Section 2

Ask anything

ChatGPT can make mistakes. Check important info.

AI Lab Overview

ChatGPT

How do I know if my clusters are meaningful or random What does a good elbow plot actually look like
Why do clusters not match real labels perfectly is my code wrong in that case

UpgradeShare

Good — these 3 questions are actually the core of K-Means understanding, not just code. I'll answer them in a way you can write in exams + understand in your head.

1. How do I know if clusters are meaningful or random?

Simple answer:

Clusters are meaningful if they show *stable structure* in the data.

You check this by:

- ✓ Silhouette score is reasonably high
 - closer to 1 = good clusters
 - around 0 = overlapping / weak clusters
- ✓ Clusters form visible groups in PCA plot
- ✓ Same types of points stay together consistently
- ✓ Clusters make real-world sense (patterns in lifestyle / obesity / etc.)

Signs clusters are basically random:

- Silhouette score near 0 or negative
- PCA plot looks like a cloud / soup
- Changing K slightly completely changes clusters
- No clear separation between groups

That means:

the data doesn't naturally cluster well OR features aren't strong enough

2. What does a good elbow plot actually look like?

Ask anything

ChatGPT can make mistakes. Check important info.

